Data Cleaning & Preprocessing Project

This project demonstrates data cleaning steps performed on the dataset 'Supplement_Sales_Weekly_Expanded.csv' using Python and Pandas in Jupyter Notebook.

Step 1: Load dataset using pandas.

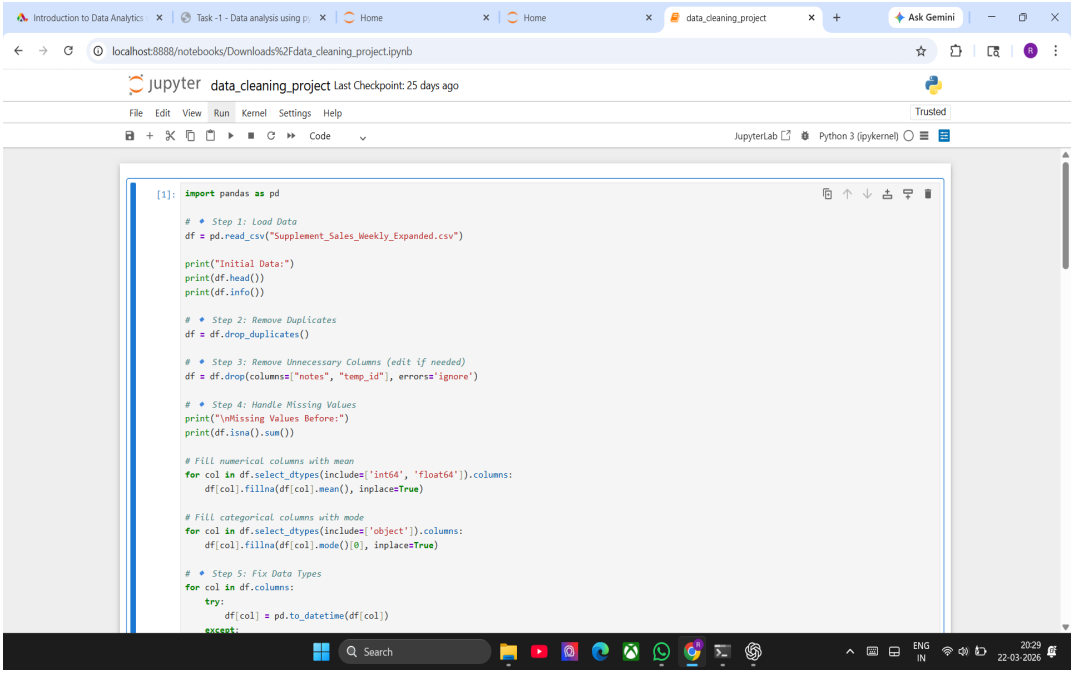
Step 2: Check data using head() and info().

Step 3: Remove duplicate records.

Step 4: Handle missing values.

Step 5: Clean text data (lowercase, remove spaces).

Step 6: Save cleaned dataset.



Introduction to Data Analytics | Task -1 - Data analysis using py | Home | Home | data_cleaning_project | Ask Gemini

localhost:8888/notebooks/Downloads%2Fdata_cleaning_project.ipynb

jupyter data_cleaning_project Last Checkpoint: 25 days ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python 3 (ipykernel)

```
    try:
        df[col] = pd.to_datetime(df[col])
    except:
        pass

# 🔹 Step 6: Clean Text Data
for col in df.select_dtypes(include=['object']).columns:
    df[col] = df[col].str.lower().str.strip()

# 🔹 Step 7: Final Check
print("\nMissing Values After:")
print(df.isna().sum())

print("\nDuplicates Left:", df.duplicated().sum())

# 🔹 Step 8: Save Clean Data
df.to_csv("cleaned_supplement_sales.csv", index=False)

print("\n✅ Cleaning Completed & File Saved!")
```

```
Initial Data:
        Date  Product Name     Category  Units Sold  Price  Revenue  \
0  2020-01-06  Whey Protein      Protein         143  31.98  4573.14
1  2020-01-06     Vitamin C      Vitamin         139  42.51  5908.89
2  2020-01-06      Fish Oil        Omega         161  12.91  2078.51
3  2020-01-06  Multivitamin      Vitamin         140  16.07  2249.80
4  2020-01-06   Pre-Workout  Performance         157  35.47  5568.79

   Discount  Units Returned Location Platform
0      0.03               2   Canada  Walmart
1      0.04               0       UK   Amazon
2      0.25               0   Canada   Amazon
3      0.08               0   Canada  Walmart
4      0.25               3   Canada    iHerb
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4384 entries, 0 to 4383
```

Search ENG IN 20:29 22-03-2026

Introduction to Data Analytics | Task -1 - Data analysis using py | Home | Home | data_cleaning_project | Ask Gemini

localhost:8888/notebooks/Downloads%2Fdata_cleaning_project.ipynb

jupyter data_cleaning_project Last Checkpoint: 25 days ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python 3 (ipykernel)

```
RangeIndex: 4384 entries, 0 to 4383
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Date            4384 non-null   object
 1   Product Name    4384 non-null   object
 2   Category        4384 non-null   object
 3   Units Sold      4384 non-null   int64
 4   Price           4384 non-null   float64
 5   Revenue         4384 non-null   float64
 6   Discount        4384 non-null   float64
 7   Units Returned  4384 non-null   int64
 8   Location        4384 non-null   object
 9   Platform        4384 non-null   object
dtypes: float64(3), int64(2), object(5)
memory usage: 342.6+ KB
None

Missing Values Before:
Date              0
Product Name      0
Category          0
Units Sold        0
Price             0
Revenue           0
Discount          0
Units Returned    0
Location          0
Platform          0
dtype: int64

Missing Values After:
Date              0
Product Name      0
Category          0
Units Sold        0
Price             0
```

Search ENG IN 20:29 22-03-2026

Introduction to Data Analytics | Task -1 - Data analysis using py | Home | Home | data_cleaning_project | Ask Gemini

localhost:8888/notebooks/Downloads%2Fdata_cleaning_project.ipynb

jupyter data_cleaning_project Last Checkpoint: 25 days ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel)

```
 6   Discount        4384 non-null   float64
 7   Units Returned  4384 non-null   int64
 8   Location        4384 non-null   object
 9   Platform        4384 non-null   object
dtypes: float64(3), int64(2), object(5)
memory usage: 342.6+ KB
None

Missing Values Before:
Date              0
Product Name      0
Category          0
Units Sold        0
Price             0
Revenue           0
Discount          0
Units Returned    0
Location          0
Platform          0
dtype: int64

Missing Values After:
Date              0
Product Name      0
Category          0
Units Sold        0
Price             0
Revenue           0
Discount          0
Units Returned    0
Location          0
Platform          0
dtype: int64

Duplicates Left: 0

✅ Cleaning Completed & File Saved!
```

Search

ENG IN 20:29 22-03-2026